

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
Distributed System
BCA — Semester 6 — 2024 — Model Answer Sheet
Subject Code: CACS352

Note: This is a model answer sheet for study purposes. Answers are based on standard curriculum topics.

Group B

Attempt any SIX questions. [6×5=30]

2. Define distributed system. Explain the goals and challenges of distributed systems.

Answer: A **Distributed System** is a collection of independent computers that appears to its users as a single coherent system. **Goals:** Resource sharing, openness, concurrency, scalability, fault tolerance, transparency.

Challenges: Heterogeneity (different hardware/OS), security, scalability (handling growth), failure handling, concurrency control, transparency (hiding distribution from users).

3. Differentiate between tightly coupled and loosely coupled systems with examples.

Answer: Tightly Coupled: Processors share memory and clock. High communication speed, single OS. Examples: Multiprocessor systems, parallel computers. **Loosely Coupled:** Processors have local memory, communicate via messages. Independent OS per node, high fault tolerance. Examples: Computer networks, distributed systems, cluster computing.

4. Explain the client-server architecture with its advantages and disadvantages.

Answer: Client-Server Architecture: Clients request services; servers provide services. **Advantages:** Centralized control, easy maintenance, security at server level, scalable by adding servers. **Disadvantages:** Single point of failure (server), network dependency, server bottleneck, higher cost for powerful servers.

Variations: Two-tier (direct client-server), Three-tier (adds application server), N-tier (multiple specialized servers).

5. What is remote procedure call (RPC)? Explain the working mechanism of RPC.

Answer: RPC allows a program to execute a procedure on a remote server as if it were local. **Working:** 1) Client calls local stub procedure. 2) Stub marshals parameters into a message. 3) Message sent to server via network. 4) Server stub unmarshals parameters. 5) Server executes the procedure. 6) Result sent back using reverse process. **Issues:** Network failures, different architectures, parameter passing semantics.

6. Describe the Bully algorithm for leader election in distributed systems.

Answer: Bully Algorithm: Used to elect a coordinator/leader when the current leader fails. **Process:** 1) Any process detecting leader failure starts an election. 2) It sends ELECTION messages to all processes with higher IDs. 3) If no higher process responds, it becomes leader and sends COORDINATOR message. 4) If a higher process responds, it takes over the election. 5) Highest-ID surviving process always wins. **Complexity:** $O(n^2)$

messages in worst case.

7. Explain distributed deadlock detection. Differentiate between centralized and distributed approaches.

Answer: Distributed Deadlock: Occurs when processes in different nodes wait for each other. **Centralized:** One node maintains global wait-for graph. Simple but single point of failure. **Distributed:** Each site maintains local wait-for graph. Sites exchange information to detect global cycles. **Approaches:** Path pushing (send paths to neighbors), Edge chasing (probe messages traverse edges), Diffusion computation (query messages spread through system).

8. What is consensus in distributed systems? Explain the Paxos algorithm briefly.

Answer: Consensus: All non-faulty nodes must agree on a single value. Required for leader election, distributed transactions. **Paxos:** A consensus algorithm that tolerates crashes. **Roles:** Proposer (suggests values), Acceptor (votes on values), Learner (learns chosen value). **Phases:** 1) Prepare: Proposer sends proposal number. 2) Promise: Acceptors promise not to accept lower proposals. 3) Accept: Proposer sends value. 4) Accepted: Acceptors vote. Majority acceptance = consensus.

Group C

Attempt any TWO questions. [2×10=20]

9. Explain the Byzantine Generals Problem. How does it relate to fault tolerance in distributed systems? Discuss practical solutions.

Answer: Byzantine Generals Problem: Loyal generals must agree on a common plan despite traitors sending conflicting information. In distributed systems, this means reaching consensus despite faulty/malicious nodes. **Relation to fault tolerance:** System must function correctly even when some components fail arbitrarily (Byzantine failures). **Solutions: PBFT (Practical Byzantine Fault Tolerance):** Requires $3f+1$ nodes to tolerate f faults. Uses multi-round voting. **Blockchain:** Proof of Work/Stake achieves consensus in open networks. **Cryptography:** Digital signatures prevent message forgery. **Limitation:** PBFT doesn't scale well; used in permissioned blockchains like Hyperledger.

10. What is distributed transaction management? Explain the two-phase commit (2PC) and three-phase commit (3PC) protocols with diagrams.

Answer: Distributed Transaction: A transaction that accesses data at multiple sites. All sites must agree to commit or abort. **2PC: Phase 1 (Voting):** Coordinator asks all participants to prepare. Participants vote YES (ready) or NO (abort). **Phase 2 (Decision):** If all YES, coordinator sends COMMIT. Otherwise ABORT. **Problem:** Coordinator failure causes blocking. **3PC:** Adds pre-commit phase. **Phase 1:** Can you prepare? **Phase 2:** Pre-commit (participants acknowledge). **Phase 3:** Commit/abort. **Advantage:** Non-blocking on single coordinator failure. **Disadvantage:** More messages, more complexity.

11. Describe Google File System (GFS) or Hadoop Distributed File System (HDFS). Explain its architecture, components, and working.

Answer: HDFS (Hadoop Distributed File System): Designed for storing very large files reliably and streaming them at high bandwidth. **Architecture: NameNode (Master):** Manages file system namespace, regulates access, maintains metadata. **DataNodes (Slaves):** Store actual data blocks, serve read/write requests, report block list to NameNode. **Working:** 1) File split into blocks (default 128MB). 2) Blocks replicated across DataNodes (default 3 replicas). 3) Client contacts NameNode for metadata. 4) Client directly reads/writes from DataNodes. 5) NameNode monitors health via heartbeats. **Features:** Fault tolerance, high throughput, write-once-read-many, rack-aware replication.

These model answers are prepared for educational purposes. Refer to textbooks for comprehensive study.